

DBMS

PL-SQL Lab 1

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1. Do all the questions discussed in the class

1. Write a function to add 2 integers values and return sum

```
5  --Q1--
6  create or replace function sumint(a int, b int)
7    returns int as
8  $$
9    Declare
10     c int;
11  Begin
12     c =a+b;
13     return c;
14  End;
15  $$
16  Language 'plpgsql';
17
18  select sumint(44, 2)
```

Data Output Messages Notifications

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	sumint	integer
1		46

2. Write a function that accepts an integer value and returns odd or even.

```
22  --Q2--
23  create or replace function check_even_odd(a int)
24    returns varchar as
25  $$
26  Begin
27    if a % 2 =0 then
28      return 'even';
29    else
30      return 'odd';
31    End if;
32  End;
33  $$
34  Language 'plpgsql';
35
36  select check_even_odd(44)
```

Data Output Messages Notifications

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	check_even_odd	character varying
1		even

Create tables Dept and Emp.

	dept_no [PK] integer	dname character varying (50)
1	10	CS
2	20	EEE
3	30	MEE
4	40	AI

	eno [PK] integer	ename character varying (50)	sal numeric (10,2)	dno integer
1	1	Pavithra	50000.00	10
2	2	Swetha	55000.00	20
3	3	Resh	54000.00	30
4	4	Kashi	60000.00	40
5	5	Deva	60000.00	10
6	6	Keerthana	40000.00	20
7	7	Nira	40000.00	10

1. Write a function that accepts an employee number and returns the salary of that employee.

```

26  --1--
27  create or replace function emp_sal(emp_no Emp.eno%type)
28      returns Emp.sal%type as
29  $$
30      Declare
31          salary Emp.sal%type;
32  Begin
33      select sal into salary from Emp where eno=emp_no;
34      return salary;
35  End;
36  $$
37  Language 'plpgsql';
38
39  select emp_sal(2)
40

```

Data Output Messages Notifications

	emp_sal numeric
1	55000.00

2. Write a function that accepts a depno and finds the no. of employees in that department.

```

44  --2--
45  create or replace function no_of_employees(mydep_no Emp.dno%type)
46  returns int as
47  $$
48      Declare
49          countemp int;
50  Begin
51      select count(eno) into countemp from Emp where mydep_no=dno;
52      return countemp;
53  End;
54  $$
55  Language 'plpgsql';
56
57  select no_of_employees(10);

```

Data Output Messages Notifications

	no_of_employees	integer
1	3	

3. Write a function that accepts an employee number and give that employee 10% high in salary if average salary of the corresponding department is greater than employees current salary.

```

59  --3--
60  create or replace function increase_sal(emp_no Emp.eno%type)
61  returns varchar as
62  $$
63      Declare
64          salary Emp.sal%type;
65  Begin
66      select sal into salary from Emp where eno=emp_no;
67      if (salary < (select avg(sal) from Emp where dno= (select dno from Emp where eno=emp_no))) then
68          update Emp set sal = sal + (sal*(0.10)) where eno=emp_no;
69      End if;
70      return 'salary updated';
71  End;
72  $$
73  Language 'plpgsql';
74  select increase_sal(3);
75

```

Data Output Messages Notifications

	increase_sal	character varying
1	salary updated	

4. Write a function that accepts an employee name and returns the dep name of the employee (assuming all names are unique)

```

77  --4--
78  create or replace function dep_name(empname Emp.ename%type)
79  returns Dept.dname%type as
80  $$
81      Declare
82          mydeptname Dept.dname%type;
83  Begin
84      select D.dname into mydeptname from Emp E join Dept D on E.dno = D.dept_no where E.ename= empname
85      return mydeptname;
86  End;
87  $$
88  Language 'plpgsql';
89
90  select dep_name('Pavithra');

```

Data Output Messages Notifications

	dep_name	character varying
1	CS	

2. Consider the following schema .

Emp(eno, ename, sal)

EMP_PROJ(eno, pno, prjt_hrs).

	eno [PK] integer	ename character varying (50)	sal numeric (10,2)	dno integer
1	1	Pavithra	50000.00	10
2	2	Swetha	55000.00	20
3	3	Resh	54000.00	30
4	4	Kashi	60000.00	40
5	5	Deva	60000.00	10
6	6	Keerthana	40000.00	20
7	7	Nira	40000.00	10

	eno integer	pno integer	prjt_hrs numeric (10,2)
1	1	101	12.50
2	1	102	15.00
3	2	103	20.00
4	3	104	18.00
5	4	105	10.00
6	5	106	22.00
7	6	107	16.50
8	7	108	14.00
9	7	109	19.00

a) Write a function ProjectLoad that returns the total project working hours for the given eno.

```

27  --a--
28  create or replace function projectload(emp_no emp.eno%type)
29      returns numeric as
30  $$
31      Declare
32          total_hrs numeric;
33      Begin
34          select coalesce(sum(prjt_hrs)) into total_hrs from emp_proj where eno = emp_no;
35          return total_hrs;
36      end;
37  $$
38  language 'plpgsql';
39
40  select projectload(1);

```

Data Output Messages Notifications

projectload	numeric
1	27.50

b) Write a PL/SQL code to update the salary of an employee if the employee earn less than the average salary. New salary is current sal + difference between current sal and average salary.

```

43  --b--
44  create or replace function update_sal(emp_no emp.eno%type)
45  returns varchar as
46  $$
47      Declare
48          avg_sal numeric;
49          curr_sal numeric;
50          new_sal numeric;
51  Begin
52      select sal into curr_sal from emp where eno = emp_no;
53      select avg(sal) into avg_sal from emp;
54      if curr_sal < avg_sal then
55          update emp
56          set sal = sal + (avg_sal - curr_sal)
57          where eno = emp_no;
58      end if;
59      select sal into new_sal from emp where eno = emp_no;
60
61      return new_sal;
62  end;
63  $$
64  language 'plpgsql';
65
66  select update_sal(3);

```

Data Output Messages Notifications

	update_sal character varying
1	54000.00

3. Consider the following tables Item(ino, iname, unit_price) Transaction(tr_no,ino,qty)

Write a function to accept an item no from the user. If transaction has been made for less than 2 times for that item(check from transaction table), delete the item from the Item table.

	ino [PK] integer	iname character varying (50)	unit_price numeric (10,2)
1	101	Puffs	10.00
2	102	Burger	50.00
3	103	Pizza	5.00
4	104	Roll	8.00
5	105	Custard	25.00

	tr_no [PK] integer	ino integer	qty integer
1	1	101	5
2	2	101	3
3	3	102	10
4	4	103	2
5	5	103	4
6	6	104	1

```

36 create or replace function delete_item(ino_input item.ino%type)
37 returns varchar as
38 $$
39 Declare
40     trans_count int;
41 Begin
42     select count(*) into trans_count from transaction where ino = ino_input;
43     if trans_count < 2 then
44         delete from item where ino = ino_input;
45         return 'item deleted';
46     else
47         return 'item not deleted';
48     end if;
49 end;
50 $$
51 language 'plpgsql';
52
53 select delete_item(101);

```

Data Output Messages Notifications

SQL
delete_item character varying
1 item not deleted

4. Consider a Bank database which includes the following tables. ACCOUNTS(ac_no,br_no, cust_no,ac_type,bal) BRANCHES(br_no, br_name, loc) CUSTOMER(cno, cname, c_type)

	cno [PK] character varying (10)	cname character varying (50)	c_type character varying (10)
1	C1	PAVITHRA	A
2	C2	SWETHA	B
3	C3	RESH	A
4	C4	KASHI	B

	br_no [PK] character varying (10)	br_name character varying (50)	loc character varying (50)
1	BR1	MAIN BRANCH	CHENNAI
2	BR2	CITY BRANCH	BANGALORE
3	BR3	TOWN BRANCH	MUMBAI

	ac_no [PK] character varying (10)	br_no character varying (10)	cust_no character varying (10)	ac_type character varying (20)	bal numeric (12,2)
1	A1	BR1	C1	SAVINGS	60000.00
2	A2	BR1	C2	CURRENT	30000.00
3	A3	BR2	C3	SAVINGS	75000.00
4	A4	BR2	C1	CURRENT	20000.00
5	A5	BR3	C4	SAVINGS	15000.00
6	A6	BR3	C2	CURRENT	50000.00

a) Write a function that accepts a threshold value and a customer number . The program updates the c_type based on the threshold value. Ie. If balance> threshold then class A, else class B.

```

51  --a--
52  create or replace function update_custtype(threshold numeric, cust_id customer.cno%type)
53  returns varchar as
54  $$
55      Declare
56          total_bal numeric;
57  Begin
58      select sum(bal) into total_bal from accounts where cust_no = cust_id;
59
60      if total_bal > threshold then
61          update customer set c_type = 'A' where cno = cust_id;
62      else
63          update customer set c_type = 'B' where cno = cust_id;
64      end if;
65      return 'customer type updated';
66  end;
67  $$
68  language 'plpgsql';
69  select update_custtype(50000.00, 'C1')

```

Data Output Messages Notifications

update_custtype
character varying

1 customer type updat...

b) Write a function called CloseBranch that takes two arguments (the branch to be closed and the branch to take over the accounts) and transfers all accounts at the closing branch to the new branch and removes the closing branch.

```

78  --b--
79  create or replace function closebranch(close_br branches.br_no%type, new_br branches.br_no%type)
80  returns varchar as
81  $$
82      Declare
83          cnt int;
84  Begin
85      update accounts set br_no = new_br where br_no = close_br;
86      delete from branches where br_no = close_br;
87
88      return 'branch closed and accounts transferred';
89  end;
90  $$
91  language plpgsql;
92  select closebranch('BR1', 'BR3')

```

Data Output Messages Notifications

closebranch
character varying

1 branch closed and accounts transferr...

c) Write a function that implements a "safe" withdrawal operation, that only permits a withdraw if there sufficient funds in the account to cover it.

```

100  --c--
101  create or replace function safe_withdraw(accno accounts.ac_no%type, amt numeric)
102  returns varchar as
103  $$
104      Declare
105          curr_bal numeric;
106  ✓   Begin
107          select bal into curr_bal from accounts where ac_no = accno;
108
109  ✓       if curr_bal >= amt then
110              update accounts set bal = bal - amt where ac_no = accno;
111              return 'withdrawal successful';
112          else
113              return 'insufficient funds - withdrawal denied';
114          end if;
115      end;
116  $$
117  language plpgsql;
118  select safe_withdraw('A1', 30000)

```

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safe_withdraw	
character varying	
1	withdrawal successf...